

AI Internship Roadmap

1. WEEK 1–2: Python + Data Foundations

Topics:

- Python basics: Loops, OOps, functions,
- Libraries: NumPy, Pandas, Matplotlib, Seaborn

Hands on Tasks:

- Data cleaning (missing values, duplicates)
- EDA (exploratory data analysis)
- Outliers handling

2. WEEK 3,4 and 5: Machine Learning (Core)

Topics:

- Supervised learning:
 - Linear Regression
 - Logistic Regression
 - Decision Trees
 - Random Forest
 - XG-Boost
- Unsupervised learning:
 - Clustering: K-Means, DBSCAN
 - Dimensionality Reduction: PCA
- Metrics:
 - Accuracy, Precision, Recall, F1, ROC AUC curve, R-square, adjusted R-square
- Train-test split, overfitting, underfitting, Trade off

3. WEEK 6 and 7 : Deep Learning Basics

Topics:

- Neural Networks fundamentals
- Activation functions
- Backpropagation (intuition)
- CNN
- RNN

Hands-on Task:

- Build ANN from scratch
- Train simple model

4. WEEK 8 and 9: NLP (Natural Language Processing)

Topics:

- Text preprocessing:
 - Tokenization
 - Stop words
 - Lemmatization
- Vectorization:
 - Bag of Words
 - TF-IDF
 - Word2Vec

Tools: Spacy, NLTK

Hands-on Tasks: Spam detection, Sentiment Analysis.

5. WEEK 10: Generative AI (GenAI)

Topics:

- What are LLMs
- Transformer architecture
- Prompt engineering
- Embeddings
- Vector databases

Tools: Langchain, Faiss, Chroma

6. WEEK 11 and 12: Final Project

Develop something based on the learning and get the exposure of industry level projects.

Note: For learning you may go through the Krish Naik and campus x tutorials on youtube.